SECURE LOCATION TRACKING FOR INTERNET OF THINGS

Capstone Project Report

MID SEMESTER EVALUATION

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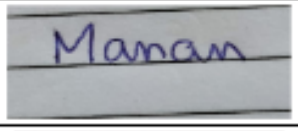
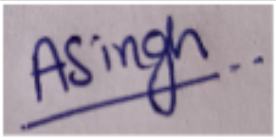
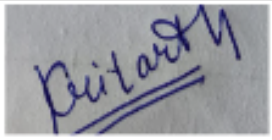
ABSTRACT

Recently, indoor localization has witnessed an increase in interest, due to the potential wide range of use in different applications, such as Internet of Things (IoT). It is also providing a solution for the absence of Global Positioning System (GPS) signals inside buildings. Different techniques have been used for performing the indoor localization, such as sensors and wireless technologies. In this paper, an indoor localization and object tracking system is proposed based on WiFi transmission technique. It is done by distributing different WiFi sources around the building to read the data of the tracked objects. This is to measure the distance between the WiFi receiver and the object to allocate and track it efficiently. The test results show that the proposed system is working in an efficient way with low cost.

DECLARATION

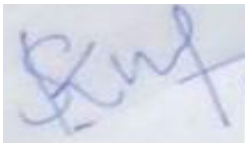
We hereby declare that the design principles and working prototype model of the project entitled Secure Location Tracking for IoT using Blockchain is an authentic record of our own work carried out in the Computer Science and Engineering Department, TIET, Patiala, under the guidance of Dr. Surjit Singh during 6th and 7th semester (2020).

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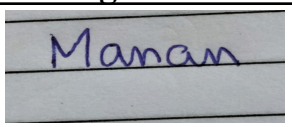
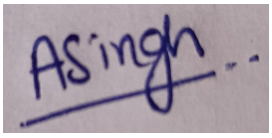
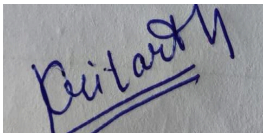
We would like to express our thanks to our mentor Dr. Surjit Singh. He has been of great help in our venture, and an indispensable resource of technical knowledge. He is truly an amazing mentor to have.

We are also thankful to Dr. Shalini Batra, Head, Computer Science and Engineering Department, entire faculty and staff of Computer Science and Engineering Department, and also our friends who devoted their valuable time and helped us in all possible ways towards successful completion of this project. We thank all those who have contributed either directly or indirectly towards this project.

Lastly, we would also like to thank our families for their unyielding love and encouragement.

They always wanted the best for us and we admire their determination and sacrifice.

Date: 26th August 2022

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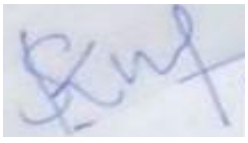


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INTRODUCTION

1.1 Project Overview

The Internet of Things (IoT) has become one of the buzzwords in Information Technology. It is an ecosystem of technologies monitoring the status of physical objects, capturing significant data, and communicating that information to software applications through IP networks. The IoT incorporates multiple long-range, short-range, and personal area wireless networks and technologies into the designs of IoT applications. It integrates many technologies, such as Internet, Zigbee, Bluetooth, infrared, WiFi, GPRS, 3G, etc, which can enable different ways to obtain the location information of various objects.

Location-based service is a primary service of the IOT. Localisation in indoor positioning systems plays an important role in the IoT. Location Based IoT applications range from tracking objects and people in real-time, assets management, agriculture, assisted monitoring technologies for healthcare, and smart homes, to name a few. Many localization algorithms and systems have been developed by means of wireless sensor networks for both indoor and outdoor environments.

The interconnection of location information of these devices brings convenience to people's lives, but there is also the problem of leaking location privacy. Therefore, effective management of the location information of the device is crucial to the development of the Internet of Things. In view of such problems existing in the traditional encryption method in the monitoring of device location information, through this project we design and verify the location information storage system of Internet of things devices based on secure storage based on the research of device location of Internet of things and the key technology of secure storage (like block chain).

The easy access and availability of wireless technologies and mobile computing and internet have led to new opportunities in developing mobile applications which purpose is to make people's life easier. Nowadays, a person can possess more than one mobile device intended for different usage such as communication, entertainment, and office work.

The Global Positioning system is the most prominent contribution in determining the position of a user and in routing him to his destination. This system uses satellites to triangulate the location of the GPS device. Though this system has

made a good impression in terms of accuracy and is the preferred location-based system for outdoor positioning, when it comes to indoor environments, GPS has proved to be inefficient. The reason for its inefficiency is that in order for GPS to perform a triangulation, the device needs to be in line-of-sight from the satellites. Moreover, the GPS system has a low precision which makes it not suitable for indoor areas. Therefore, when it comes to indoor positioning systems, other alternatives such as Bluetooth, WIFI, RFID and Infrared Red are more preferable.

At present, buildings are equipped with Wireless Local Area Network (WLAN) access points, such as airports, supermarkets, hospitals, museums, etc. It can be effectively implemented to use these access points as a source and to receive WiFi signals to locate user location in these internal environments. However, when it comes to localization of objects indoors, here we need to know more accurate information about the position of such objects inside a building.
There are various indoor localization algorithms based on wireless signals. The main challenges in using these algorithms are to provide acceptable accuracy and deployment cost. For these reasons, a lot of interest has been gained in many research communities using WiFi for indoor localization techniques. Therefore, the proposed method of this paper uses WiFi signals strength to compute the distance between the user's device and the WiFi sender (Access Points (AP) or router). On the other hand, the problem is that WiFi signals are changing as the obstacles, such as walls, equipment found in the building, and the people around it can affect the signal strengths. The proposed indoor localization technology based on WiFi signal solves this problem by distributing the access point to guarantee the coverage of the whole building. It is simple in implementation and estimation and it can localize the position of a user's device on each floor.

1.2 Need Analysis

With the rapid development in wireless technologies and the Internet, the Internet of Things (IoT) is envisioned to be an integral part of our daily lives. Localization-based services are among the most attractive applications related to the IoT. They are actually, thanks to the deployment of networks of sensors, able to collect and transmit data in order to determine the target's position. A plethora of localization systems are proposed in the literature. These localization systems are based on different positioning approaches, different techniques and different technologies, making them appropriate for some applications and inappropriate for other applications

The position estimation is an essential component of wireless sensor networks applications [1], especially in the environmental monitoring applications such as [2]: animal dwelling place monitoring, water quality monitoring and farming precision. In addition, the location information enables various emerging applications such as inventory management, intrusion detection, road traffic tracking, health monitoring, etc. [2].

The Internet of Things (IoT) is a paradigm shift in the enhancement of the connectivity of the Internet. It provides the technical feasibility for a huge number of sensors, actuators and smart devices to remain connected with it. It adds several emerging services which are essential for connected living initiatives. Among those value added services, localization is one of the prominent one. Localization has a lot of importance in the connected living environment such as security, surveillance and maintenance of law and order. Localization of moving objects leads to tracking which is essential for similar applications. Both localization and tracking are required for emerging applications in several sectors.

Today the students mark the attendance offline on the sheets and there is huge consumption of papers on a daily basis in the colleges.

1.3 Research Gaps

a. Indoor Localization System Using Wi-Fi Technology:
In this paper, the Wi-Fi signal strength-based object detection and tracking system in the indoor environment was proposed. The proposed system used the WiFi signal strength read from the distributed access points to evaluate the position of the required object. The location is assigned using the MAC address of the smartphone of each user involved. The strength of reading WiFi signals from an access point refers to the location of such a device. The smartphone device is connected to the nearest access point based on the WiFi signal strength, and it is disconnected from the far points. The obtained results showed the accurate performance of the proposed system in terms of allocating objects.
It has some limitations like security issues with no security of the stored data and proper management of it.

b. Wi-Fi-based Indoor Positioning System
The proposed system works in two phases; offline and online phase. The offline phase involves creating a radio map. The radio map stores distributions of RSS values from all detected APs at specific points which are known as marking positions. The marking positions together with the MAC address of each detected APs and their corresponding RSS values will be stored in the database to create the radio map. In the online phase, when the user initializes the mobile application, the sniffer component will collect samples of RSS values from all detected APs. To estimate the position of the user, the data collected will be compared in the radio map. This has limitations like the accuracy of the system can be increased by applying various techniques such as backtracking which will avoid the system from choosing between two close measurements. The system that is proposed and developed in this major project is still in the prototype stage.

c. Location of Things(LoT) : A Review and Taxonomy of Sensors Localization in IoT Infrastructure
In this paper, a recent extensive analysis of localization techniques and hierarchical taxonomy and their applications in different contexts is presented. This taxonomy of the localization technique is classified based on the presence of offline training in localization, namely self-determining and training dependent approaches. The future of localization and its implementation for IoT is manifested. Recent localization technologies are analysed and a taxonomical model is presented based on IoT Infrastructure. The emphasis of classification is done based on the presence of offline training state in the algorithms. Various Comparison tables are given for

different approaches. How everything can be localized and its futuristic application of IoT-based localization is justified. Different noise distribution and their effect on localization are exposed. Thus the use of localization has different approaches for different scenarios. In this the system has not been implemented and only the efficiency and analysis of various models are done.

d. SECURITY AND PRIVACY PRESERVING OF SENSOR DATA LOCALIZATION BASED ON INTERNET OF THINGS

The paper addresses the problem of locating the data sensed from the various applications into the cloud, optimizing the energy consumption of storing the data into the cloud and eluding the illegal access by, proper placement of the conflicting and the non-conflicting in the separate host. This process of data locating is based on the non-dominant sorting genetic algorithm-II to arrive at an optimal solution to maximize the quality of service by minimizing the energy consumption, maximizing the resource utilization and enhancing the security by protecting the privacy of the data set by storing the *INDs* and the *NDS* in different hosts. The further evaluation using the network simulator-II, provides with the efficiency of the proposed method of data locating compared to the prevailing methods. The paper does not provide the optimization providing the optimal solution reducing the access time of the virtual machines.

1.4 Problem Definition and Scope

The problem of Wi-Fi based indoor localization of a device consists in determining the position of client devices with respect to access points. Many techniques exist to accomplish this, and these may be classified into four main types: *received signal strength indication* (RSSI), fingerprinting, angle of arrival (AoA) and time of flight (ToF) based techniques.

In most cases the first step to determine a device's position is to determine the distance between the target client device and a few access points. With the known distances between the target device and access points, trilateration algorithms may be used to determine the relative position of the target device, using the known position of access points as a reference. Alternatively, the angle of arriving signals at a target client device can be employed to determine the device's location based on triangulation algorithms.

A combination of these techniques may be used to increase the accuracy of the system.

1.5 Assumptions and Constraints

Assumptions

1. The user device has Wifi service available and always on.
2. One user can have only one account to access the service.
3. The admin can make unlimited requests and change roles, permissions etc.

Constraints

1. The device must reside within the specified region of Localization.
2. The accuracy of the system can be increased by using more Access Points.
3. The development project must be completed within 6 months.

1.6 Standards

- Laptop
    - Model : Inspiron 5490
        - Processor: Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz 2.11 GHz
        - RAM: 8.00 GB (7.79 GB usable)
        - System type: 64-bit operating system, x64-based processor
        - Display adapter:
            - NVIDIA GeForce MX250
            - Intel(R) UHD Graphics
- Google colab:
  Colab notebooks execute code on Google's cloud servers, meaning we can leverage the power of Google hardware, including GPUs and TPUs, regardless of the specifications of the laptop.
  On cloud server allocated RAM is 12.69 GB RAM and 107.72 GB Disk storage

1.7 Approved Objectives

Localization is utilized in various fields to encourage certain activities e.g security, service provision, navigation and monitoring and tracking. Since the approximate position of objects is the primary thought localization, the objectives talked about give more information about localization.

- *Service providing*: Patient care in hospitals, security and maintenance of corporate houses, food services in restaurants and hotels are some of the services provided. Security service provision in homes and industries and tracking services to the users are used in static based because sensors are fixed. Technologies such as GPS, infrared, Internet, WiFi, satellite, Bluetooth all contribute to the process of localization to provide services.

- *Navigation*: It is mainly mobility dependent. Localization in territories, and areas using communication among sensor nodes combined with other techniques has made it possible to navigate and collect data. Exchange of information between various systems likewise information can be exchanged by the use of IoT equipped vehicles and the main system facilitates easy navigation.

- *Security*: Major and significant difficulties of IoT is security which may be overcome by some secure storage techniques like storing the information on cloud networks. Any device equipped with this system should be able to store the information securely.

1.8 Methodology

In RSSI-based localization systems, the accuracy of the final location largely depends on the range. Therefore, in order to obtain higher accuracy, we need to try various possible methods to improve the range accuracy. In equation (3) we get the relationship of RSSI and distance, where the parameter n means the exponential attenuation factor to distance. In the indoor environment, the external factors deeply affect range value; even in the same location of a node, the value of n may be variable with different circumstances. In order to maintain a certain location accuracy, adjusting the value of n is necessary. Here a method is designed to adjust the range value between blind node and beacon, which depends on RSSI and real distance of beacons. As shown in Figure 5, the real distance between blind node M and beacon node B1 is dm, the calculated distance is m d ′ . The real distance between beacon nodes B1 and B2 is d, while the calculated distance is d′ .

RSSI = A - 10nlgd (3)
RSSI = A - 10n'lgd'

Therefore, we have: logd n = n' x $\log_d$ The initial value of path loss factor is specified as n′ , and the real distance d can be calculated according to the coordinator of two beacon nodes. d′ is the distance calculated last time.

1.9 Project Outcomes and Deliverables

The proposed solution offers an embodiment to measure and evaluate the fulfillment of objectives. The embodiments to be offered includes:

1. Real time localization system using IoT.
2. Different methods would be analyzed for the said tasks and most ambient one would be incorporated.
3. The user data would be safeguarded using different security protocols and to provide further security it would be stored at decentralized databases using the blockchain technology.
4. We would try to come up with techniques to scale up the demonstrated ideas.
5. Emphasis would also be laid to incorporate a number of other functionalities to make the system diverse.

1.10 Novelty of Work

1. No comprehensive work dedicated to secure Location tracking for IoT using Blockchain has been done.
2. Indeed at present general research in this area is still in its infancy,
3. However, work has been carried out to use various algorithms to improve the accuracy but previous works have not comprehensively considered XGboost and Ensemble algorithm for the further improvement and its secure storage using Blockchain

2.1 Literature Survey

2.1.1 Theory Associated With Problem Area

The Internet of things conceptualised a big connectivity model where all devices and services from all the places connected to the internet all time. The 'things' of IoT are heterogeneous devices consisting of sensors and actuators. The sensor senses signals from the environment and the actuator acts on the environment. The IoT opens an important dimension called 'Location of Things' (LoT) where things get geographical position information. There are billions of 'things' in IoT and continuously adding. To organise this big data from huge amounts of heterogeneous devices the location information plays an important role. IoT pulls data from huge devices, filters out and organises them by means of location information. Hence location acts as a search engine for organising big data and devices.

The localization principle developed 20 years back with the development of GPS for military application further it is commercialised. Currently all device connected to the internet uses location information e.g. Google Map, Uber, Waze, Foursquare etc. Hence the location based services (LBS) comes into play as a paradigm shift from traditional localization i.e GPS. GPS is just a subset of LBS since it cannot localize 'things' (sensor and actuators) always. There are a huge number of scenarios like indoor network, underwater network, unfavourable weather conditions when GPS do not work. So alternative techniques are required. Hence development of a huge number of techniques and protocols is done for these scenarios and localization plays an important role for services in IoT infrastructure.

A major Location based interactive model of IoT (showing IoT-Cloud-WSN integration) part of IoT infrastructure consists of heterogeneous sensors so it can be called as sensing service on demand based on the user. IoT infrastructure is a paradigm shift from traditional sensor networks as shown in the figure. It consists of a cloud or virtual sink which schedules and controls physical sensor networks based on location of users. Traditional WSN has grown into Sensor Cloud architecture in IoT infrastructure. Here efficient sensor networks are used for Sensing as a Service (SeaaS) model of the IoT infrastructure.

The architecture of location based interactive models of IoT is presented in the figure below.

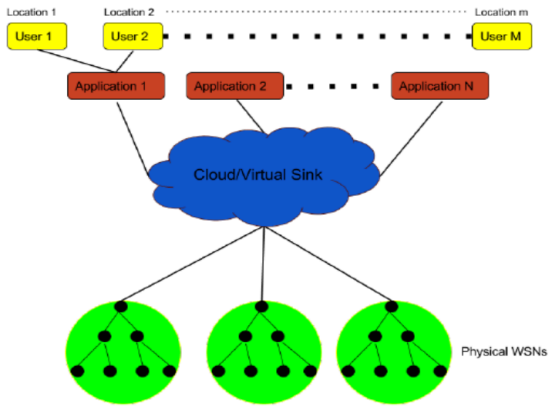


Fig2.1 Location based interactive model of Iot

2.1.2 Existing Systems and Solutions

From the existing literature and IoT application it is found that there is a need for review of existing technology to address the localization problem to the IoT infrastructure. The selection of proper localization methods in the IoT scenario aims to address the restriction and avoid the pitfalls. The proposed taxonomy classifies the localization approach based on offline training. Since IoT infrastructure analyses huge data and devices. There is a huge scope of application offline training for localization. So the training step is the key for the taxonomic hierarchy of localization methods in IoT infrastructure.
The localization approach is reviewed into two category considering IoT scenario as :
(1) Self Determining Method (2) Training Dependent Method

A. Self Determining Method:
Self-determining methods can be subdivided into four categories namely :
(1) Geometric Method. (2) Mobility model (3) Path planning (4) Statistical Approximation. The geometric method considers various geometric parameters to estimate location. This method checks triangular information like lateration, angulation or connectivity among sensor nodes. This method is categorised based on connectivity, centroid calculation, landmark design, region overlap, clustering etc. Mobility models take advantage of the mobility pattern of anchor nodes of the network . The mobility pattern can be random or specified. This is further classified on the basis of mobility patterns like random walk, random direction, self-organisation etc. Path planning is an advanced version of anchor mobility model in which anchor nodes move in a specified path in the network to localize the whole network. Few scenario
of these categories like static path model and dynamic path model have huge applications in robotic sensor and actuator networks. Statistical approximation techniques use standard approximation methods like Least Square, Maximum Likelihood, Multidimensional Scaling, Semidefinite Programming etc to estimate location.

B. Training Dependent Method
This method uses an offline training phase before the online localization step. These are classified into (1) Fingerprinting (2) Stochastic Model (3) Machine Learning approach. Fingerprinting method is further classified into signal, visual, motion and hybrid types of fingerprints. Stochastic Models can be of POMDP (Partially Observable Markov decision process), MDP (Markov decision process), SG (Stochastic Game), HMDP (Hierarchical Markov decision process) and CMDP (Constrained Markov decision process). Machine learning based approach is classified into LMFF (Levenberg- Marquardt Feed-Forward), RBFN (Radial Basis Function Network), BRGD (Bayesian Regularization Gradient Descent), RPML (Resilient Back-propagation Multilayer) and MLP (Multi-Layer Perceptron).
This approach of localization exploits geometric info to find location.

2.1.3 Sample Table for Literature Survey

S. No.	Roll Number	Name	Paper Title	Findings
1	101903311	Manan Ghai	IoT based localization and tracking	In this work, we present the current improvements in localization and tracking using IoT. We start with the differences between the IoT based localization and the non-IoT based systems. Then we show the suitability of IoT for localization and tracking. We show several current applications of IoT in localization and tracking.
			Indoor Localization System Using Wi-Fi Technology	In this paper, the Wi-Fi signal strength-based object detection and tracking system in the indoor environment was proposed. The proposed system used the WiFi signal strength read from the distributed access points to evaluate the position of the required object. The location is assigned using the MAC address of the smartphone of each user involved. The strength of reading WiFi signals from an access point refers to the location of such a device. The smartphone device is connected to the nearest access point based on the WiFi signal strength, and it is disconnected from the far points.
2	101903315	Akshat Singh	Location of Things (LoT): A Review and Taxonomy of Sensors Localization in IoT Infrastructure	In this paper, a recent extensive analysis of localization techniques and hierarchical taxonomy and their applications in different contexts is presented. This taxonomy of the localization technique is classified based on the presence of offline training in localization, namely self-determining and training dependent approaches.
			Wi-Fi-based Indoor Positioning System	The proposed system works in two phases; offline and online phase. The offline phase involves creating a radio map. The radio map stores distributions of RSS values from all detected APs at specific points which are known as marking positions. The marking positions together with the MAC address of each

				detected APs and their corresponding RSS values will be stored in the database to create the radio map. In the online phase, when the user initializes the mobile application, the sniffer component will collect samples of RSS values from all detected APs. To estimate the position of the user, the data collected will be compared in the radio map.
3	101903321	Kritarth Gupta	SECURITY AND PRIVACY PRESERVING OF SENSOR DATA LOCALIZATION BASED ON INTERNET OF THINGS	The paper addresses the problem of locating the data sensed from the various applications into the cloud, optimizing the energy consumption of storing the data into the cloud and eluding the illegal access by, proper placement of the conflicting and the non-conflicting in the separate hosts.
			A Novel Approach to Data Storage Using Blockchain Technology	The proposed system currently carries some limitations like the adapter server query data integrity maintenance as the queries itself are not managed on the blockchain. For the initial engine proposed in this paper, naive data structures were used which can end up being computationally expensive in the long run.

2.1.4 Problem Identified

The problem we are addressing is the huge data sensed during localisation on IoT devices which will act as a medium to increase the efficiency of networks in the global world along with secure storage of data which is a major concern because such information can be more susceptible to various attacks. Thus in order to eliminate this, we would like to propose an ideology through this project which is to verify whether data coming from sensors is authenticated or not and is stored securely.

2.2 Software Requirement Specifications

2.2.1 Introduction

2.2.1.1 Purpose

To overcome inefficient work processes, there are various software systems available to speed up the attendance process and reduce manual work. An online attendance management or digital attendance platform is one of them, which is developed to automate daily attendance. Additionally, it helps to maintain accurate records and generate summarized student attendance reports.

The purpose of this management system is to help teachers track student's attendance, absentee record, attendance history & other related documents and keep track of the lectures/labs attended or missed by students. It is the system used to document the time at which students mark their attendance, student's name, roll number, course code and whether they are attending lecture or labs and at which particular venue. Attendance management is done by recording all the above said fields, using our App and thereby recording all the information on spreadsheets on daily or monthly basis, which can be easily accessed by the professors later on for the record.

2.2.1.2 Intended Audience and Reading Suggestions

The intended audience is the engineering class on whom it will be implemented and testing the Attendance Management System. Also, the document is to be utilized by the engineering professor to evaluate the software's design and features.

This document is to be read by the developing team which include 3 members, our mentor of the project and the panel evaluating faculty. The specifications of the project should be read carefully for understanding the overall working flow of the software and its further scopes.

2.2.1.3 Project Scope

This project is able to take attendance of the students and fill their details. The Attendance Management System will allow the teacher incharge to maintain a record of attendance of students in their respective classes from a PC. Also, the system will permit the teacher incharge to maintain all the essential details regarding a particular student. Furthermore, the program provides different modes to edit attendance, view statistics, and take notes on elements pertaining to attendance. The goal is to provide a professor with an easy, portable solution to attendance record maintenance and attendance statistics.

This project can further be enhanced to take attendance of faculty, staff of college to have their details filled and other profile information like salary, personal details etc.

2.2.2 Overall Description

2.2.2.1 Product Perspective

The Attendance Management System is intended to replace the manual model of attendance record keeping by means of roll call and paper records. The roll call and paper records are replaced with a single interaction between the professor and the Attendance Management System. Professors will be able to view details regarding attendance of individual students on their PC and quickly maintain attendance records.

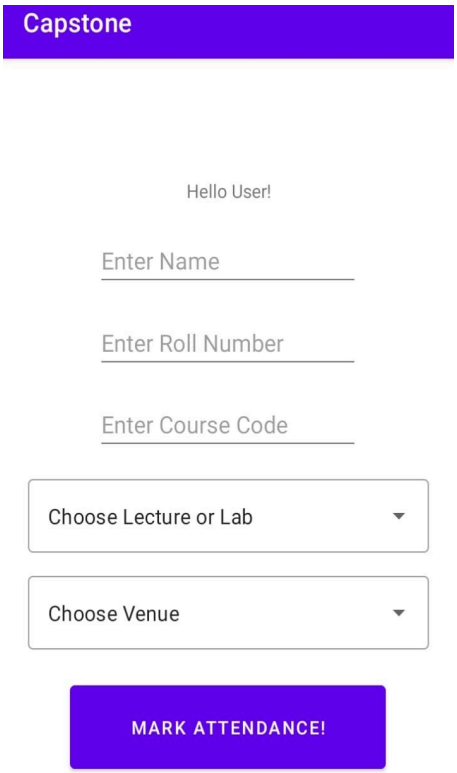
The software is developed for attendance purposes. It is using Api calls for interaction between the frontend and the backend. For frontend and styling purposes we are using Android Studio, JavaScript, ReactJS. In the backend we are using Java along with python and ML models for cleaning the clutter/unwanted data, preprocessing the data, applying feature engineering and presenting the final output in an ordered manner so that it is easily readable by the user.

2.2.2.2 Product Features

The main feature of the Management System is that it keeps a record of all essential details of each individual student which can be accessed by the Admin or faculty who can view it later or as and when required. The app will continuously store the data of the students' attendance whenever an entry is made in the form of a list/dataframe onto the google sheets database. Attendance of each student in every subject would be displayed in this list. Also, the system can further be modified to view pertinent statistics using data analysis on a student's attendance record. Finally, another feature of the Attendance Management System is its flexibility and ease of use.

2.2.3 External Interface Requirements

2.2.3.1 User Interfaces



2.2.3.2 Hardware Interfaces

1. Mobile phones and laptops with active internet connection are required.
2. The systems developed are compatible with 32-bit, 64-bit machines and with the Android Version 9.0 and higher versions.
3. Scanner application to scan the QR code for marking the attendance.

2.2.3.3 Software Interfaces

1. The software is developed for android, ios, windows 7, windows 8, windows 10, etc.
2. The JavaScript library used is ReactJS and AppScripts for the connection between frontend Application interface and the Database.

2.2.4 Other Non-functional Requirements

2.2.4.1 Performance Requirements

The performance of the product made depends on the machine learning model created and the hardware interface developed. Performance benchmarks can be set to judge the output obtained against the input provided. The output desired compared to the output obtained for a given input defines the degree of performance for our Attendance System. The System should provide the desired results for an input:

1. The verification of attendance.
2. The location of the student.

2.2.4.2 Safety Requirements

1. Proper safety measures and precautions must be incorporated for the assembly of hardware components.
2. The hardware components attached should not be opened and tampered with.
3. An adequate power supply should be provided to prevent the system from getting burned.
4. Connecting wires and other hardware materials should be handled with care and connected properly to eliminate any short circuits.

2.2.4.3 Security Requirements

1. Information developed throughout the learning process should be accessible by anyone.
2. Only the admin or the developers of the product should have access to make changes in the model and the dataset. (which consists of student details).

2.3 Cost Analysis

Components	Price	Quantity	Total Price (Rs.)
Wifi Access Point	3000	3	9000

2.4 Risk Analysis

Risk analysis is an important step for any project, to identify any unwanted scenarios for which the level of preparedness is insufficient. While we can predict and reduce the impact of the maximum number of outcomes based on our research and expertise in the subject in hope that it does not have any consequences as the project evolves, it is natural to not be able to cover all aspects of the domain.
As we analyse all the risks and possibilities which can occur while developing the project, it can help us streamline our risks and its impact, improving the chances of a successful project.
The following are some risks that the user should be aware of while using our system:

1. Our System will be unable to function without WiFi.
2. Only the rooms and labs that our system has been programmed to understand will be interpreted.
3. If the requests are longer, the response time is longer, and the system will produce the delayed results.

METHODOLOGY ADOPTED

3.1 Investigative Techniques

The technique we are using to accomplish our project's objective is experimental. With the help of ML Models, and Networking attributes we will classify the attendance of students captured by the app that will enable us to identify the correct attendance marked by the student.

3.2 Proposed Solution

Our proposed solution revolves around the idea of a comparative differentiation between the widely used Wifi localization methods based upon different parameters and its ability to be integrated with a blockchain enabled system for secure storage of the obtained location coordinates.

In the last few years research is going on for finding the efficient algorithm for localization. The localization of the sensor node is a two step process of first physical measurement with some reference and second estimate the location from the physical measurement data. Sometimes nodes with limited hardware do not conduct physical measurement rather they try to calculate the position from neighbor node information. The following approaches for localization are existing in the literature. These are based on range measurement, available from anchor nodes, fine-grained, coarse-grained, incremental and concurrent localization methods. The basis of classification of the above scheme is based on either anchor node or range measurement. From the existing literature and IoT application it is found that there is a need for review of existing technology to address the localization problem to the IoT infrastructure. The selection of proper localization methods in the IoT scenario aims to address the restriction and avoid the pitfalls. The proposed taxonomy classifies the localization approach based on offline training. Since IoT infrastructure analyzes huge data and devices. There is a huge scope of application offline training for localization. So the training step is the key for the taxonomic hierarchy of localization methods in IoT infrastructure. The classification is presented in Fig. The localization approach is reviewed into two categories considering the IoT scenario as Self Determining Method Training Dependent Method.

We would be performing a thorough analysis of the different localisation algorithms and would finally recommend the best possible solution.

From the product point of view we would be building a prototype to demonstrate our findings based on the results of the research. We would be designing an attendance based system that would require the location access of the respective users and mark their attendance based on the access points of a network they are connected to. Our proposed model matches the BSSID of the access point to which the user is connected with the list of permitted BSSID of the access points present in that classroom.

Further we would be creating a web interface where a teacher can access the attendance of the students based on their attributes like room no. , course code, batch, day, time etc. and would get the final excel files sent to their email address.

3.3 Work Breakdown Structure

Sr. No.	Activity	Month	January		February				March				April				May				June			
		Week No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1.	Identification formulation and Planning of Project	Plan																						
		Actual																						
2.	Study of utility, localization and courses.	Plan																						
		Actual																						
3.	Designing and Development	Plan																						
		Actual																						
4.	Hardware Implementation	Plan																						
		Actual																						
5.	Testing and Performance Analysis	Plan																						
		Actual																						
6.	Results Evaluation	Plan																						
		Actual																						
7.	Final Evaluation	Plan																						
		Actual																						

3.4 Tools and Technology

We would be making use of a vast variety of tools and different technologies in our project work which include:

1. Machine learning models.
2. Jupyter.
3. Python.
4. Networking Principles.
5. Blockchain.
6. IOT devices and their functionalities.
7. Android Studio.
8. Java
9. JavaScript
10. ReactJS

4.1 System Architecture

4.1.1 MVC Architecture

The MVC architecture is made up of three parts:

- **Model:** The XG Boost model that works as the model for our system. It is responsible for classifying the input attendance and then sending the result of the classified attendance to the .
- **View:** The updated display for the teacher works as the view of our system.
- **Controller:** : The display screen works as the controller for our system. It is responsible for displaying the selected students list.

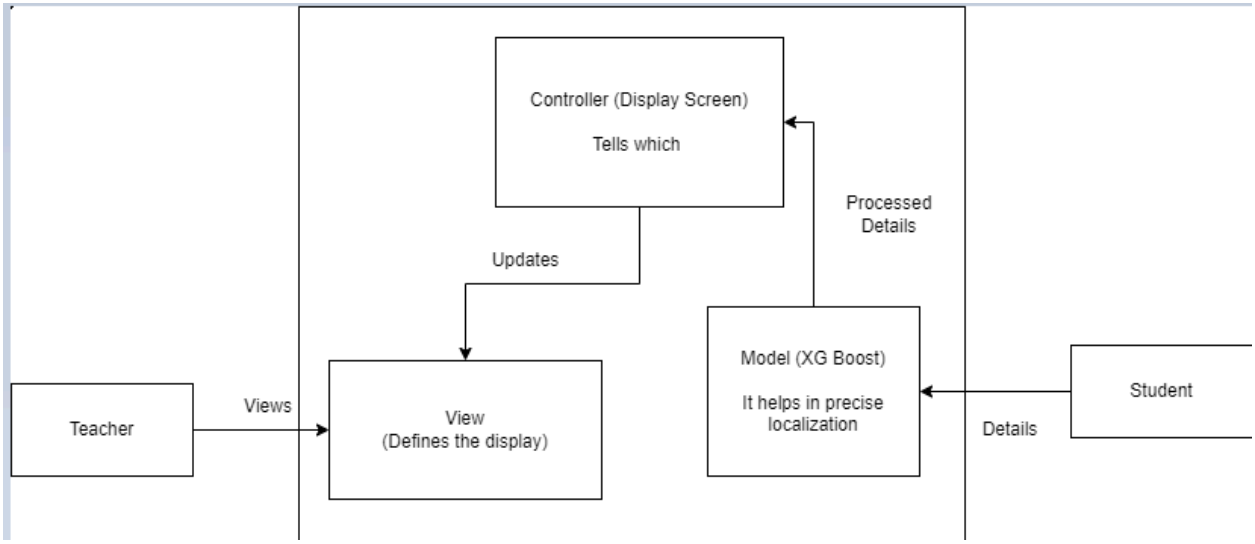


Fig (MVC Architecture)

4.1.2 Block Diagram

It is a diagram of a system where the main components or functions are represented by blocks linked together by lines that highlight the connections between the blocks.

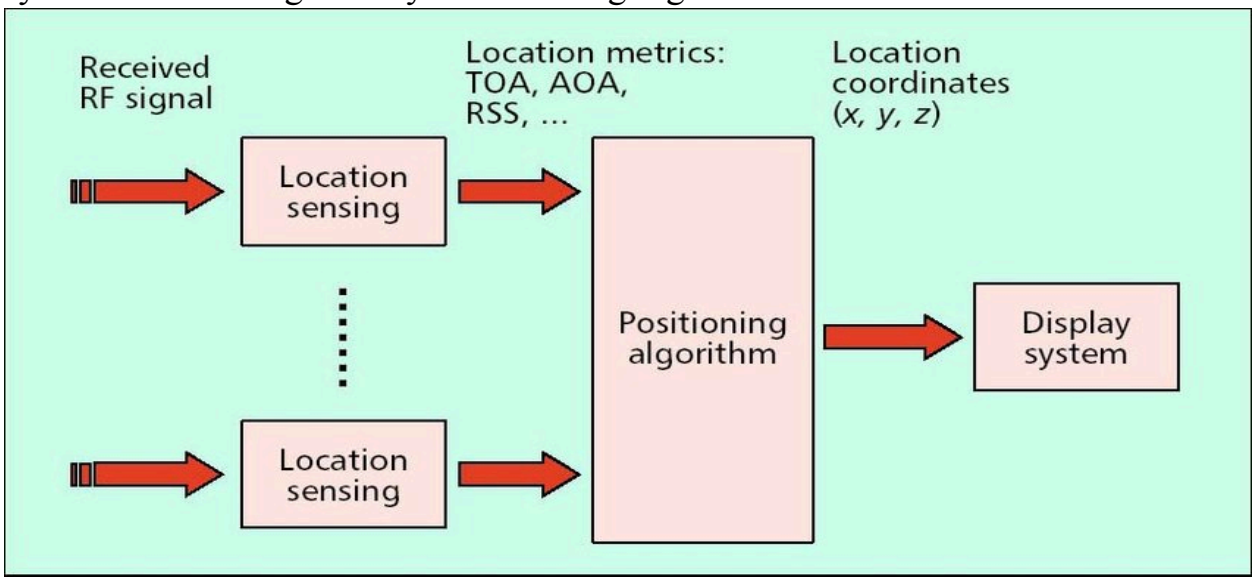
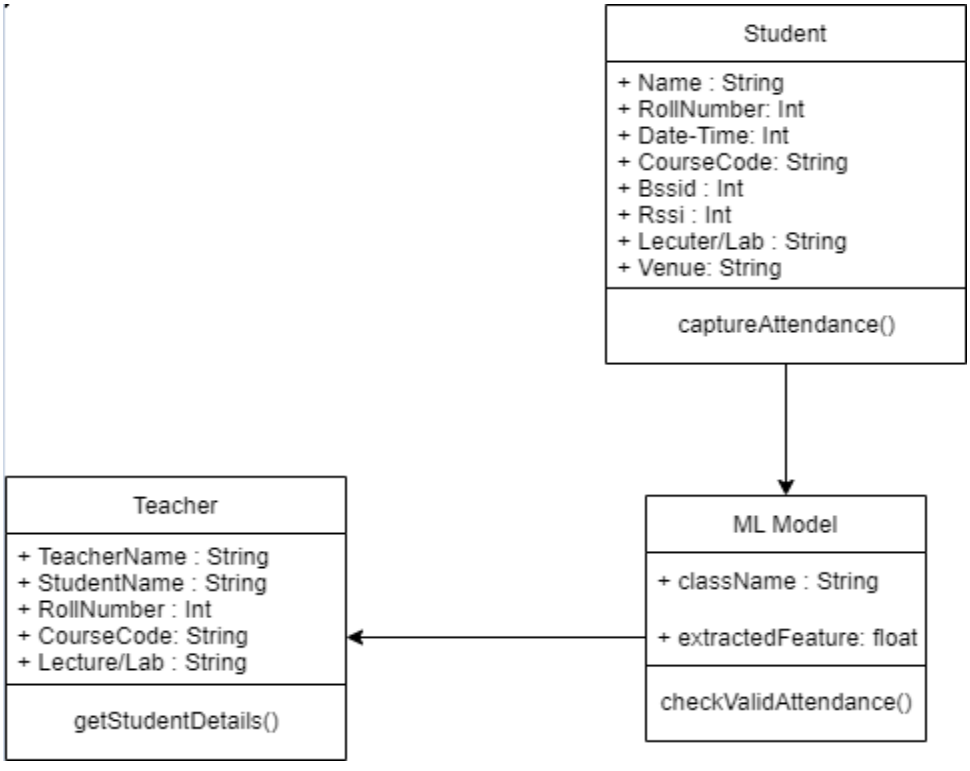


Fig 3. Block Diagram

4.2 Design level diagrams

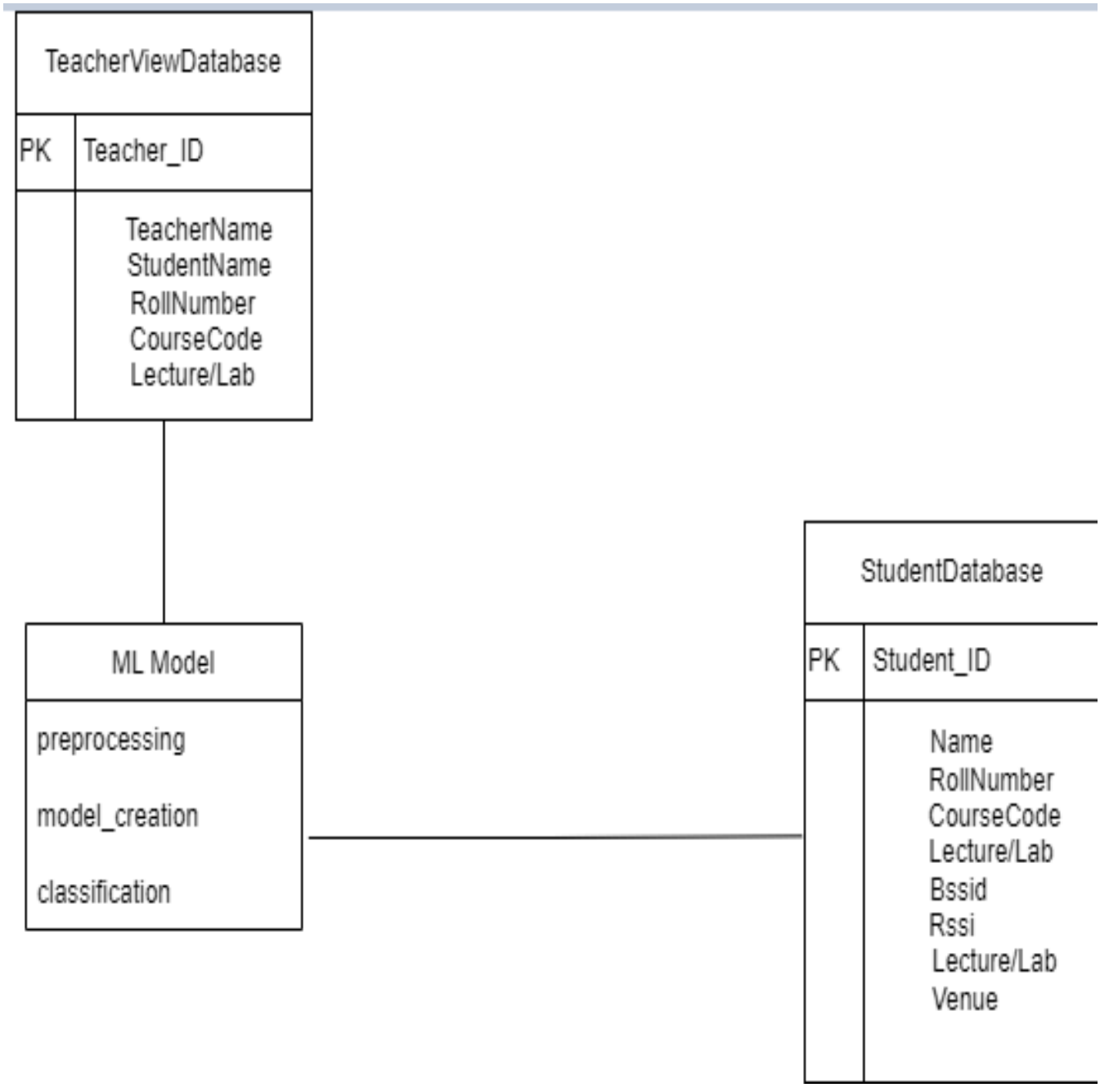
4.2.1 Class Diagram

Our class diagram has three major classes. In the first class, i.e Student class like getting student name, roll Number, Course Code etc. details from the students.



4.2.2 ER Diagram

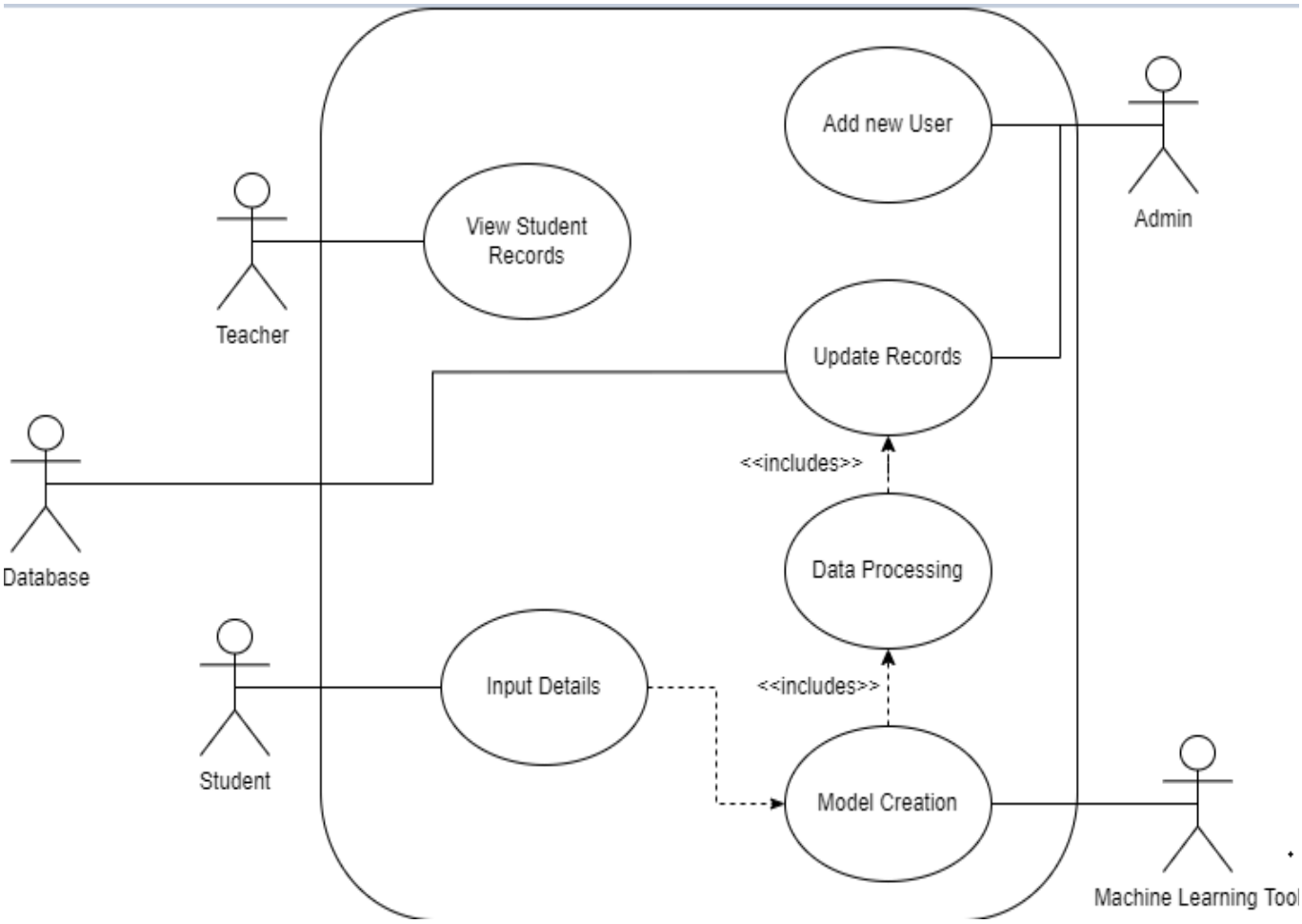
An ER Diagram is a form of flowchart that shows the relationships between "entities" like people, things, or concepts within a system. They are most frequently used in the disciplines of software engineering, business information systems, education, and research to build or troubleshoot relational databases.



4.3 User Interface Diagrams

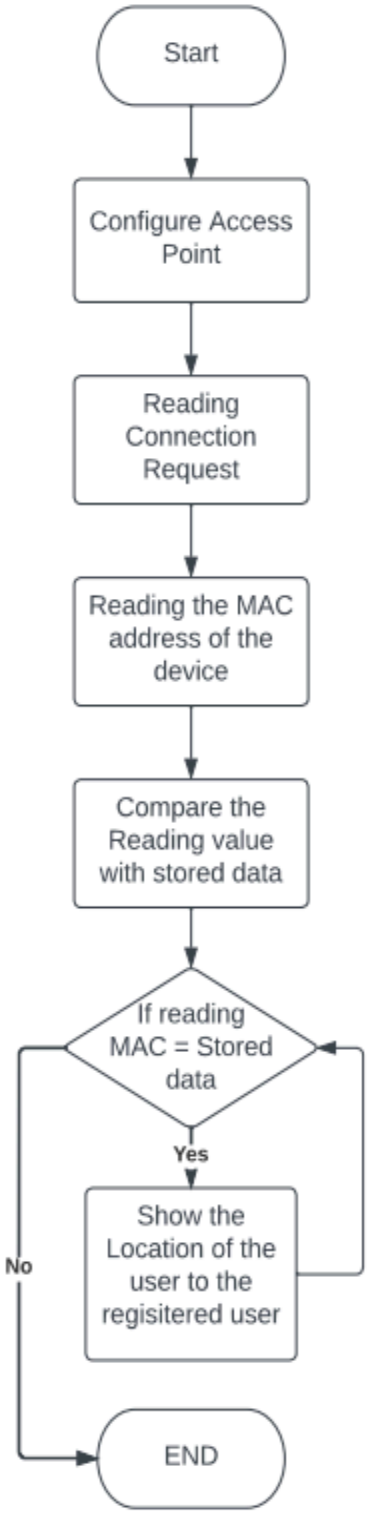
4.3.1 Use Case Diagram

A graphical representation of a user's potential interactions with a system is called a use case diagram. It displays different user types and use cases for the system.



4.3.2 Activity Diagram

An activity diagram is essentially a flowchart that shows how one activity leads to another. The activity might be referred to as a system operation. One operation leads to the next in the control flow.



4.4 Snapshots of Working Prototype

The following steps are followed for the implementation of localization prediction:

- Importing necessary Python libraries like librosa, matplotlib, pandas, tensorflow, sklearn, numpy.
- Using spectrogram and waveplot for Exploratory Data Analysis (EDA) and comparing librosa's pre-processing methods.

```
In [24]: plot_cm(x_test, y_test, model, room_encoder)
```

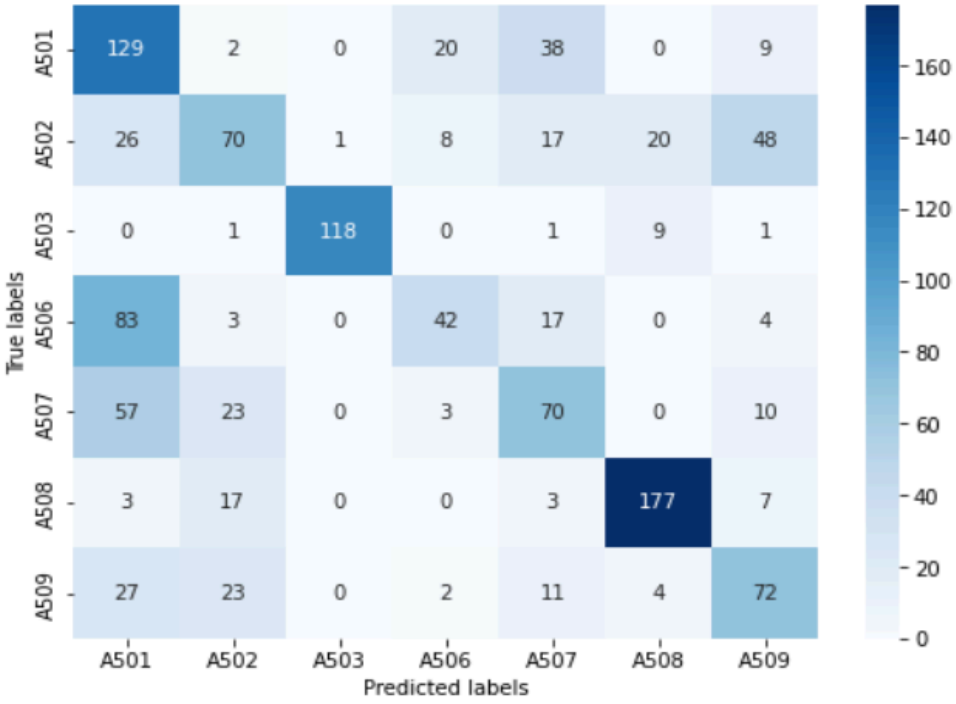
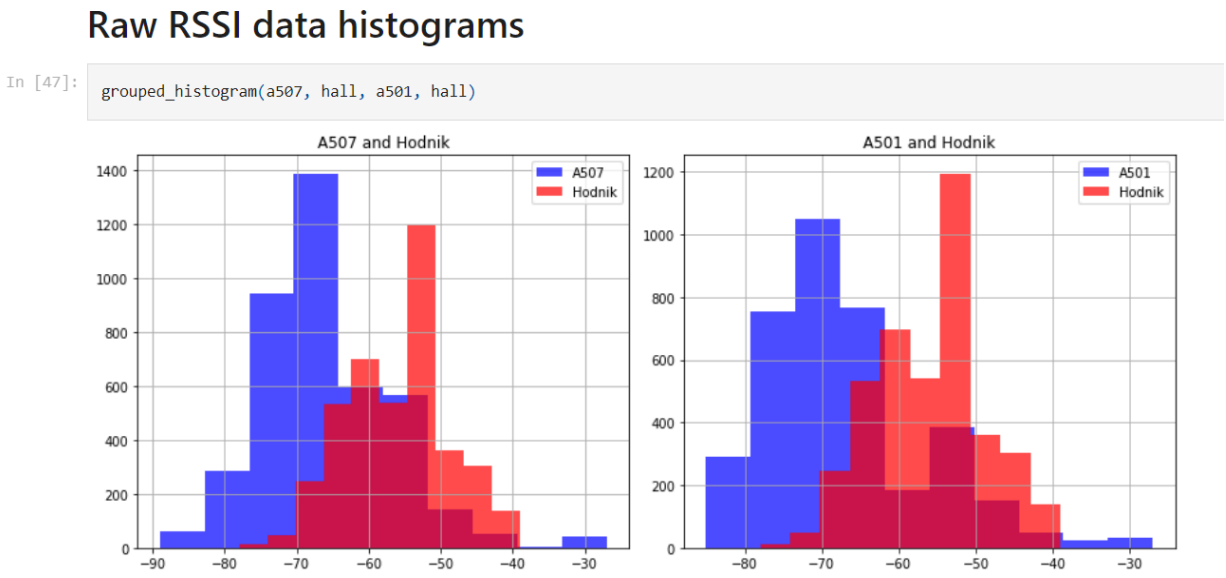


Fig4.1 heatmap of prediction using Wifi



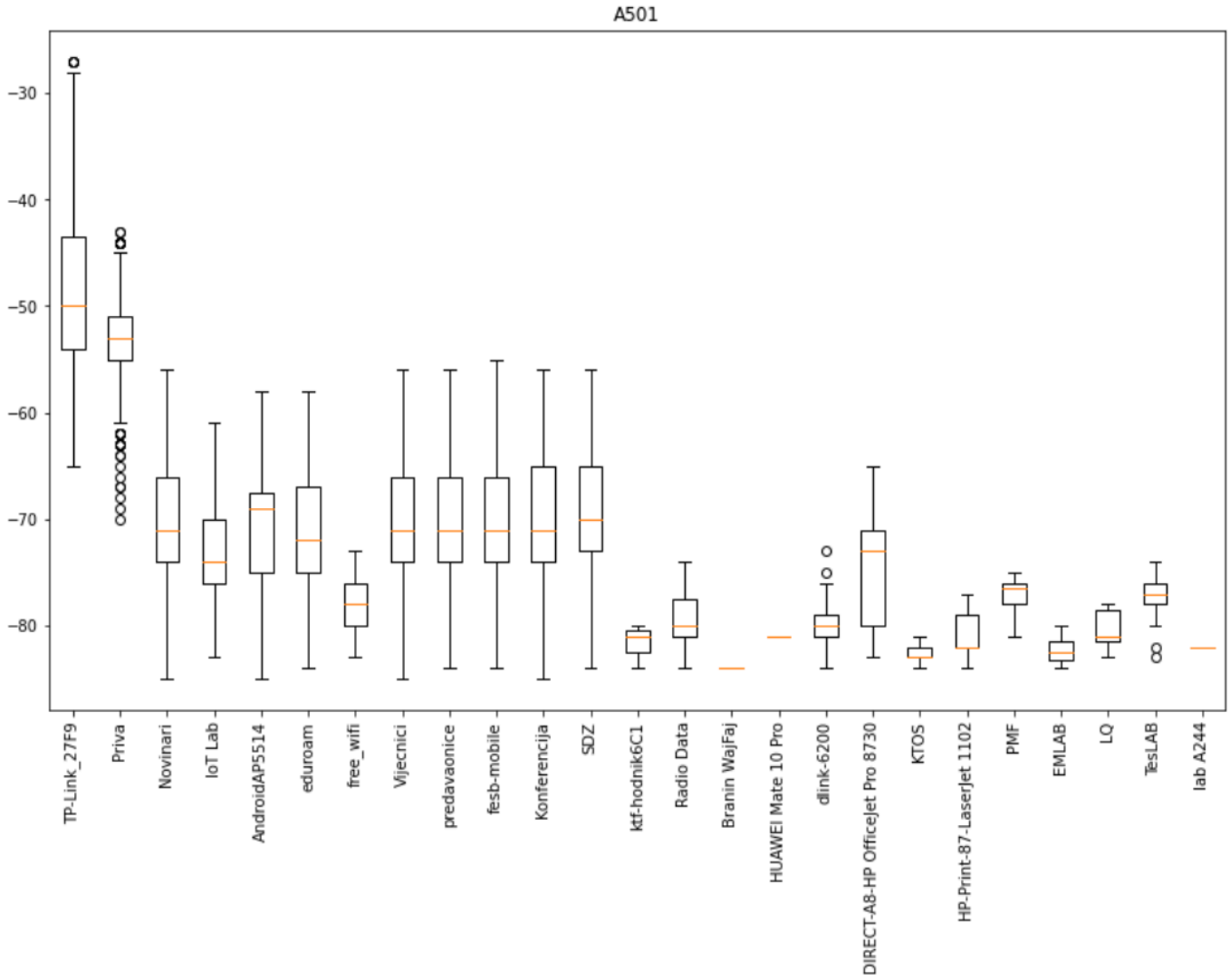


Fig 4.2 Boxplot wifi Localization

CONCLUSIONS AND FUTURE SCOPE

5.1 Work Accomplished

We started with our project doing research on the various localization methods being used up till now. We carried out a thorough comparison between the methods like ML based, Network based, calculation based etc , Comparing them on different grounds of accuracy and many other parameters. We then proposed our solution which consisted of an ensemble model for location prediction. From the security perspective we further proposed that we would be using a Blockchain based storage system instead of the conventional storage systems being used.

From the product perspective to showcase an application of the above carried research we created an Attendance Monitoring System. Which makes use of the location permissions and the BSSID of the users marking their attendance.

5.2 Conclusions

Student data collected from wifi access points across the campus can be used for marking attendance in order for smooth and secure tracking of students.

5.3 Environmental Benefits

The Environmental Benefits include:

1. Saving a large amount of paper which was being used to take the attendance.
2. Saving electricity which was earlier being consumed to keep a track of the manually marked attendance by the teachers on their systems.

5.4 Future Work Plan

We would further be creating a web interface for the teachers to directly access the attendance of the students , by logging in and using different filters as required.

We would further be employing different methods based on the application of machine learning models to further improve the results of the existing application.

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